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Defective g-C_3N_4 prepared by the $NaBH_4$ reduction for high-performance H_2 production

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KEYWORDS. Defects, graphitic carbon nitride, chemical reduction, charge separation, H_2 production

ABSTRACT. Defects play a significant part in the promoting photocatalytic activity for H_2 production. Various methods such as chemical reduction, have been performed to the metal oxide based photocatalyst, Herein, we present the $NaBH_4$ reduction route to introduce the defects into the graphitic carbon nitride (g-C_3N_4) to enhance the photocatalytic activity. A new -C≡N group is observed in the FTIR spectra of treated g-C_3N_4 nanosheets indicating the presence of structural defects. At the same time, the B signal appears in the X-ray photoelectron spectroscopy suggesting that B is doped in the g-C_3N_4 during the treatment. All these results manifested that multiple types of defects are introduced in the g-C_3N_4 during the $NaBH_4$ treatment. The UV-vis spectra illustrate that the absorption band edge of g-C_3N_4 is extended from 420 nm to 450 nm after $NaBH_4$ treatment. It proves that the band gap of g-C_3N_4 turns narrow owing to the introduction of defects. Photocatalytic H_2 production of defective g-C_3N_4 is ~5 folds better than the pristine g-C_3N_4. To understand the enhanced mechanism, the apparent quantum efficiency, photoluminescent spectra, transient photocurrent, and electrochemical impedance spectra are investigated, all these results prove that the charge separation efficiency is greatly strengthened in the defective g-C_3N_4. Upon the above, the enhancement of catalytic activity can be attributed to both the broad light adsorption range and highly efficient charge separation process.

Introduction

With the demand for the new sustainable energy, H_2 energy has been treated as a potential and clean energy fuel due to its high energy storage density without pollution emission. Solar water splitting is a prevailing route to generate H_2 by using photocatalyst under solar light irradiation.[1-3, 4] It provides an efficient way to transform the solar energy into chemical energy. However, its conversion efficiency is in a quite low level after near half a century investigation.[5] It is a great challenge for exploring high efficient photocatalyst. Based on the working mechanism of the photocatalyst,[6] it needs to solve these three critical issues: widening photo response range, improving charge separation efficiency and lowering overpotential for H_2 reduction. TiO_2 has been demonstrated to be the most common and stable photocatalyst.[2] However, it only has a response to a UV light because of its large band gap, which limits its practical application. Since black TiO_2 was reported by Chen and coworkers,[7-8] the defective TiO_2 has been extensively investigated.[9-11] TiO_2 with defects like oxygen vacancies exhibited extended visible light adsorption and efficient charge separation. Various methods like H_2 thermal or plasma treatment, Al vapor reduction, $NaBH_4$ thermal reduction, have been applied to TiO_2 to produce defective TiO_2 or other oxides.[12-16] In our previous reports, defective TiO_2 was prepared by chemical reduction of TiO_2 under mild conditions.[15, 17-18] The defects can broaden the light adsorption range and promote the charge separation process. Recently, a few groups reported the defective g-C_3N_4 prepared from different methods also exhibited the excellent photocatalytic performance.

Since 2009, Wang *et al*[19] reported the graphitic carbon nitride (g-C_3N_4), which has gained much attention due to its visible light response.[20] g-C_3N_4 has a moderate band gap of 2.7-2.9 eV, relating to the optical wavelength of 420-460 nm. However, its bandgap is obviously too large to absorb the most part of the solar spectrum. A few methods such as heteroatom doping,[21-27] amorphous,[28-29] defective[30-34] and copolymerization[35-38] have been employed to expand the light adsorption range. Comparing with TiO_2, defective g-C_3N_4 is expected to possess high photocatalytic activities. It is still a great challenge to develop a facile route to gain defective g-C_3N_4.

Herein, we demonstrated that defective g-C_3N_4 (DCN) prepared via $NaBH_4$ reduction route exhibit higher photocatalytic performance than the pristine g-C_3N_4. The X-ray photoelectron spectra illustrate that B signal appears in the DCN indicating g-C_3N_4 was doped with B. A new vibration peak at 2170 cm^{-1} was observed in the FTIR spectra, which is assigned to the vibration peak of -C≡N group. It proves the presence of -C≡N group in the DCN samples. The UV-vis spectra disclosed that the band edge red-shifts about 40 nm. The corresponding band gap was tuned from 2.94 eV to 2.71 eV. Transient photocurrent, photoluminescent spectra and impedance plot all show the better charge separation efficiency in the DCN samples. DCN samples exhibit about 4-5 folds better H_2 production rate than pristine g-C_3N_4. The H_2 production rate increase from 42 μmol/h to 222 μmol/h for 20 mg catalyst under visible light irradiation (λ>420 nm). DCN samples exhibit super stability over 4 continuous cycles photocatalytic reaction. The apparent quantum efficiency of DCN can reach ~17 % at 420 nm filter with a bandwidth of 20 nm, which is ~ 2 folds higher than that of pristine g-C_3N_4. DCN also exhibit H_2 production activity at 475 nm, however, no H_2 was detected at this wavelength. These demonstrate DCN has broad light response range and highly efficient charge separation, those are the reasons why DCN has higher photocatalytic activities.

Experimental Section

Synthesis of photocatalysts

All chemicals were directly used without further purification.

Synthesis of g-C_3N_4: Pristine g-C_3N_4 was prepared by calcination of urea in a muffle furnace. Specifically, 15 g of urea was calcined at 550 °C in a muffle furnace for 3 h with a heating rate of 1 °C/min.

Synthesis of defective-g-C_3N_4 (DCN): Firstly, 0.8 g of g-C_3N_4 and 0.6 g of $NaBH_4$ are fully ground in the mortar for 30 minutes. Then, the ground solid powder is transferred into the crucible and roasted in a 300 °C muffle furnace. In other words, the muffle will be heated up to 300 °C before roasting the mixture at 15min, 30min, 45min, 60 min. Finally, the obtained solid powder is washed two times with deionized water and ethanol, respectively. The sample is dried in 60 °C oven. The samples prepared were denoted by DCN-1, DCN-2, DCN-3 and DCN-4 for different reaction time, respectively.

Characterizations

UV-vis diffuse reflection spectroscopy (DRS) of the samples was characterized by a Shimadzu UV-2600 spectrophotometer. The X-ray diffraction (XRD) measurements were carried out on the Bruker D8 Advance X-ray diffractometer) with Cu Kα radiation (λ=0.15406 nm) as the incident beam at 40 kV and 40 mA. Scanning electron microscopy (SEM) images were obtained on the JEOL S-4800. FT-IR spectra were collected by a SHIMADZU IRAffinity-1S. The Hitachi F-7000 instrument was used for the photoluminescence (PL) measurement for the photocatalyst with an excitation wavelength of 320 nm. X-ray photoelectron spectroscopy (XPS) measurements were conducted on a Thermo Fisher ESCALAB 250xi multifunctional X-ray spectrometer. Electron paramagnetic resonance (EPR) were collected on JES-FA200 ESR Spectrometer.

Surface photovoltage spectroscopy (SPV)

The surface photovoltage spectroscopy measurement is home made of a sample chamber, a monochromatic light source with a light chopper (SR 540), and a lock-in amplifier (SR 830-DSP) for signal detection. The monochromatic light source is generated by a 300 W Xe lamp (PLS-SXE 300, Beijing Perfectlight Co. Ltd) with a monochromator (SBP500, Zolix). All samples are measured under ambient pressure at room temperature and samples are not further treated prior to the SPV measurement.

Photocatalytic activities

Photocatalytic H_2-production measurements were conducted in a 200 mL close glass reactor with an external flow of cooling water kept at 5 °C and vacuum pressure. In order to analyze photocatalytic H_2-production performance, the 20 mg catalyst was suspended in 90 mL deionized water and ultrasonic dispersed for 15 min and then poured the solution into the reactor. 10 mL TEOA was added into as a hole scavenger. The 1.0 wt% Pt cocatalyst was loaded by the photodeposition process in H_2PtCl_6 solution. The reaction solution was irradiated with UV-vis light (300 W Xe lamp) for an hour to load Pt nanoparticles onto the catalyst. The visible light source was provided by a 300 W Xe lamp with an ultraviolet cut-off filter ($\lambda > 420$ nm) for the hydrogen evolution experiments. The evolved H_2 was measurements using gas chromatography (Shimadzu GC-2014C) equipped with a thermal conductivity detector (TCD).

Wavelength-dependent H_2-production tests were conducted with the same H_2 production system. The light source is generated a 300 C Xe lamp equipped with the bandpass filter. The following equation is employed to calculate the apparent quantum efficiency (AQE) :

$$\text{AQE (\%)} = \frac{2 \times \text{number of evolved } H_2 \text{ molecules}}{\text{Number of incident photons}}$$

Results and Discussion

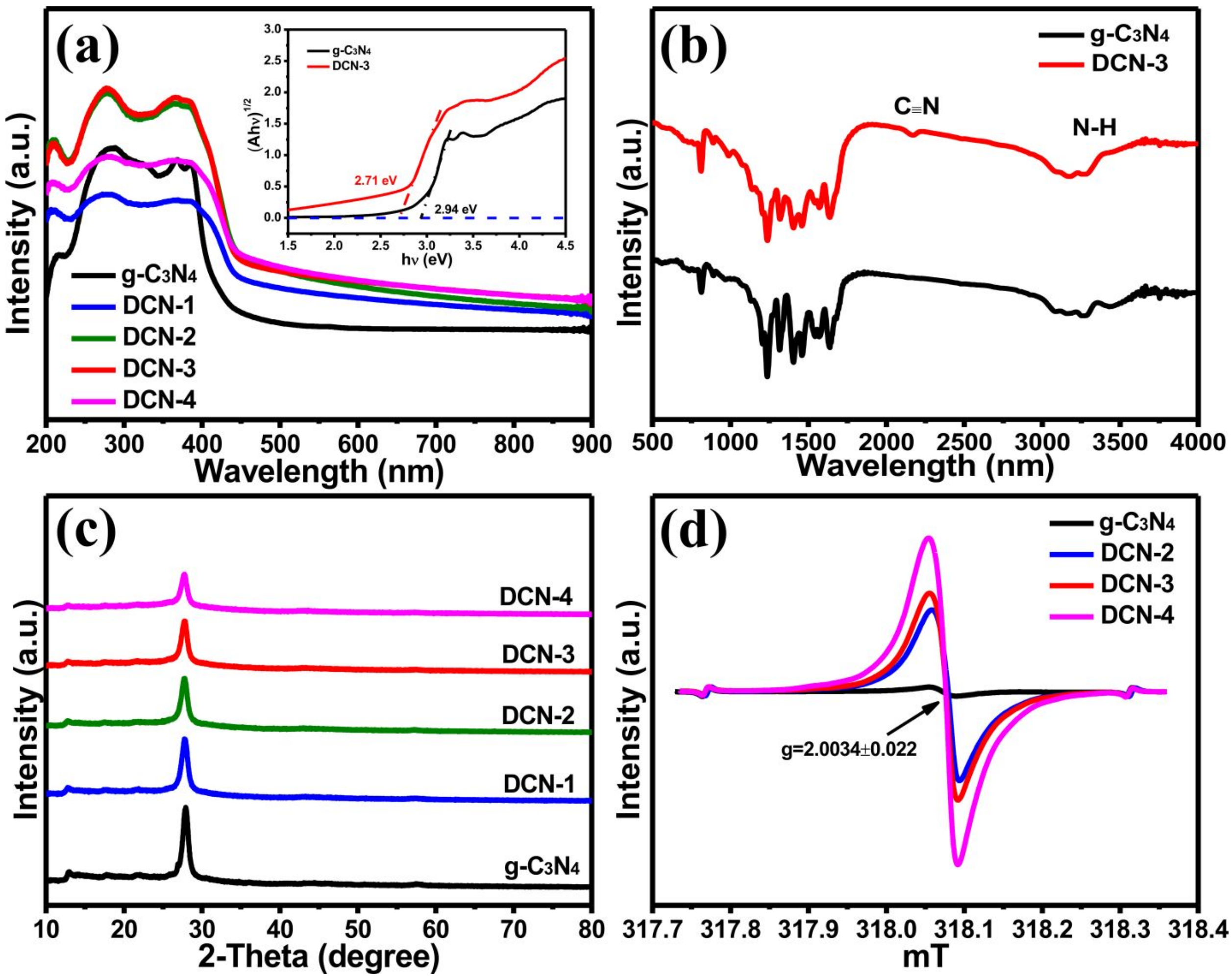

Figure 1. (a) UV-vis spectra of the pristine g-C_3N_4, DCN-1, DCN-2, DCN-3 and DCN-4. The insert graph is the Tauc plots of $(\alpha h\nu)^{1/2}$ vs photon energy for the pristine g-C_3N_4 and DCN-3. (b) FT-IR spectra of g-C_3N_4 and DCN-3. (c) XRD of g-C_3N_4, DCN-1, DCV-2, DCN-3 and DCN-4. (d) ESR of g-C_3N_4, DCV-2, DCN-3 and DCN-4.

The pristine g-C_3N_4 was prepared by the calcination of urea at 550 °C for 3 hours. $NaBH_4$ and g-C_3N_4 were uniformly mixed by ground for 30 min, and then the mixture was placed into the preheated oven at 300°C and hold for different time (15, 30, 45 and 60 minutes for DCN-1~4, respectively). Figure S1 shows the synthesis route of g-C_3N_4 and DCN samples. The sample color changes from light yellow to dark yellow. Figure 1a displays UV-vis spectra of g-C_3N_4 and DCN samples. The band edge of g-C_3N_4 gradually red-shifts from 420 nm to 450 nm, the corresponding band gap changes from 2.94 eV to 2.71 eV (Figure 1a inset). That indicates that the reduction treatment can reduce the band gap of g-C_3N_4 and broaden the light absorption range in the visible light region. Figure 1b illustrates that the FTIR spectra of g-C_3N_4 and DCN-3. There are two groups of characterization peaks, one locates at 3300-3500 cm^{-1}, attributing to the -NH and OH vibration peaks. The IR peak at 803 cm^{-1} related to the breathing mode of the tri-s-triazine units. Other 2 main peaks in the 1250–1590 cm^{-1} were assigned to the stretching mode of C−N in heterocycles. The peak at 1632 cm^{-1} was assigned to the stretching mode of C=N bond, which further proved the presence of sp^2 hybridized network.[37, 39] A new peak was observed at 2170 cm^{-1}, attributing to the -C≡N vibration peak. That indicates that the -C≡N group formed in the reduction reaction may result in some molecular structural defect in the g-C_3N_4.[31, 40-41] Figure 1c exhibit the X-ray diffraction (XRD) pattern of g-C_3N_4 and DCN samples. The diffraction peaks of pristine g-C_3N_4 appearing at 13.1° and 27.4° corresponding to the (100) and (002) plane, which are assigned to the inter-layer structural packing and the interplanar staking peaks of conjugation system, respectively.[19, 42] After $NaBH_4$ reduction treatment, the diffraction peaks keep no change. That indicates that the aggregation state of g-C_3N_4 keeps similar to the pristine g-C_3N_4. Figure 1d shows the electron paramagnetic resonance (EPR) of pure g-C_3N_4 and DCN samples measured in the dark at room temperature. An isotropic signal at near g=2.0034 with a symmetric line of nearly

Lorentzian line shape was detected, which is attributed to the conduction electrons in the localized π-states of g-C_3N_4.[43-44] The presence of the conduction electrons will be helpful for the photocatalytic reactions. The EPR intensities were enhanced significantly after the treatment of $NaBH_4$ indicating a large number of conduction electrons can be developed by the $NaBH_4$ treatment. Figure S2 shows the scanning electron microscopy images (SEM) of g-C_3N_4 and DCN samples. Those clearly illustrate that all these samples display nanosheets structure. That indicates that the $NaBH_4$ treatment does not change the morphology of g-C_3N_4.

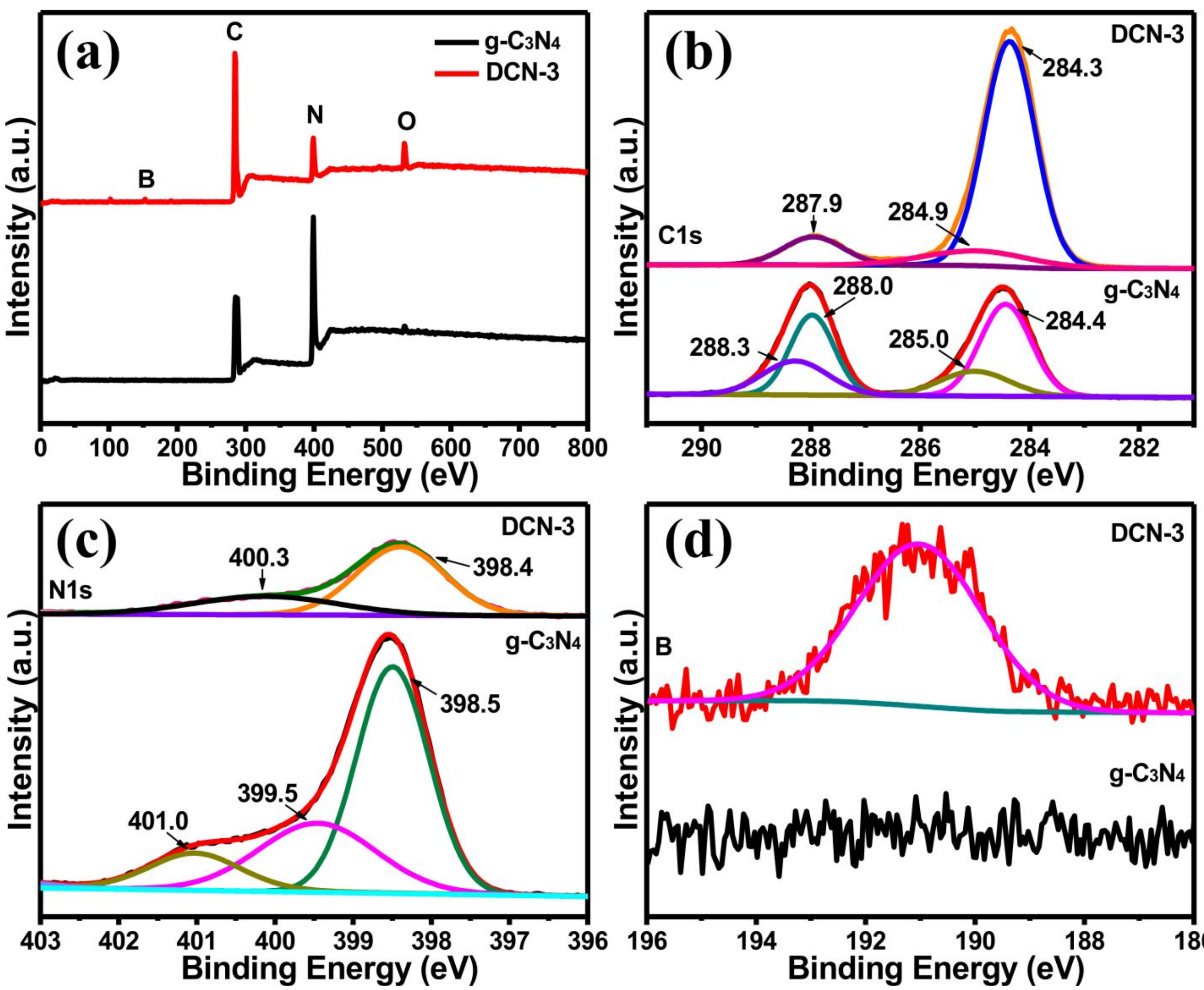

Figure 2. X-ray photoelectron spectroscopy of g-C_3N_4 and DCN-3. (a) The full XPS survey, high-resolution XPS spectra of C1s (b), N1s (c) and B1s (d).

X-ray photoelectron spectroscopy (XPS) analyses were carried out to characterize the chemical composition of g-C_3N_4 and DCN-3. The full XPS survey of g-C_3N_4 (Figure 2a) displays three peaks at 284, 400 and 532 eV corresponding to the C, N and O elements, respectively. A weak peak at 190 eV was observed in the DCN indicating the existence of the B in the sample. Figure 2b-2d are the high-resolution XPS spectra of C1s, N1s and B1s of DCN. Four main peaks with binding energies at 284.4 eV, 285.0 eV 288.0 and 288.3 eV can be found on the C1s core-level XPS spectrum of g-C_3N_4, which are assigned to sp^2 C=C/C-C, sp^3 C-C/C-N, sp^2 bond carbon (N-C=N) and carboxyl O=C-OH, respectively.[45] The C1s XPS spectrum of DCN can be divided into three peaks at 284.4, 285.0 and 287.9 eV, respectively. That indicates carboxyl can be removed by the reduction reaction. The high-resolution N1s XPS spectrum of g-C_3N_4 contains three peaks at 398.5 eV, 399.5 eV and 401.0 eV, respectively, which is corresponding to the sp^2 bonded N (C-N=N), graphite N (N-C3) and amino groups (C-NH/C-NH_2). Only two kinds of N was observed in the DCN sample, one is sp^2 N (C-N=C, 398.4 eV) and amino group N (C-NH and C-NH_2, 400.3 eV). After treatment, the oxygen content clearly increases due to the thermal treatment carried out in the air environment. No B signal is observed in the g-C_3N_4, it appears in the DCN sample. That indicates the B was introduced into the g-C_3N_4 and form B doped g-C_3N_4.

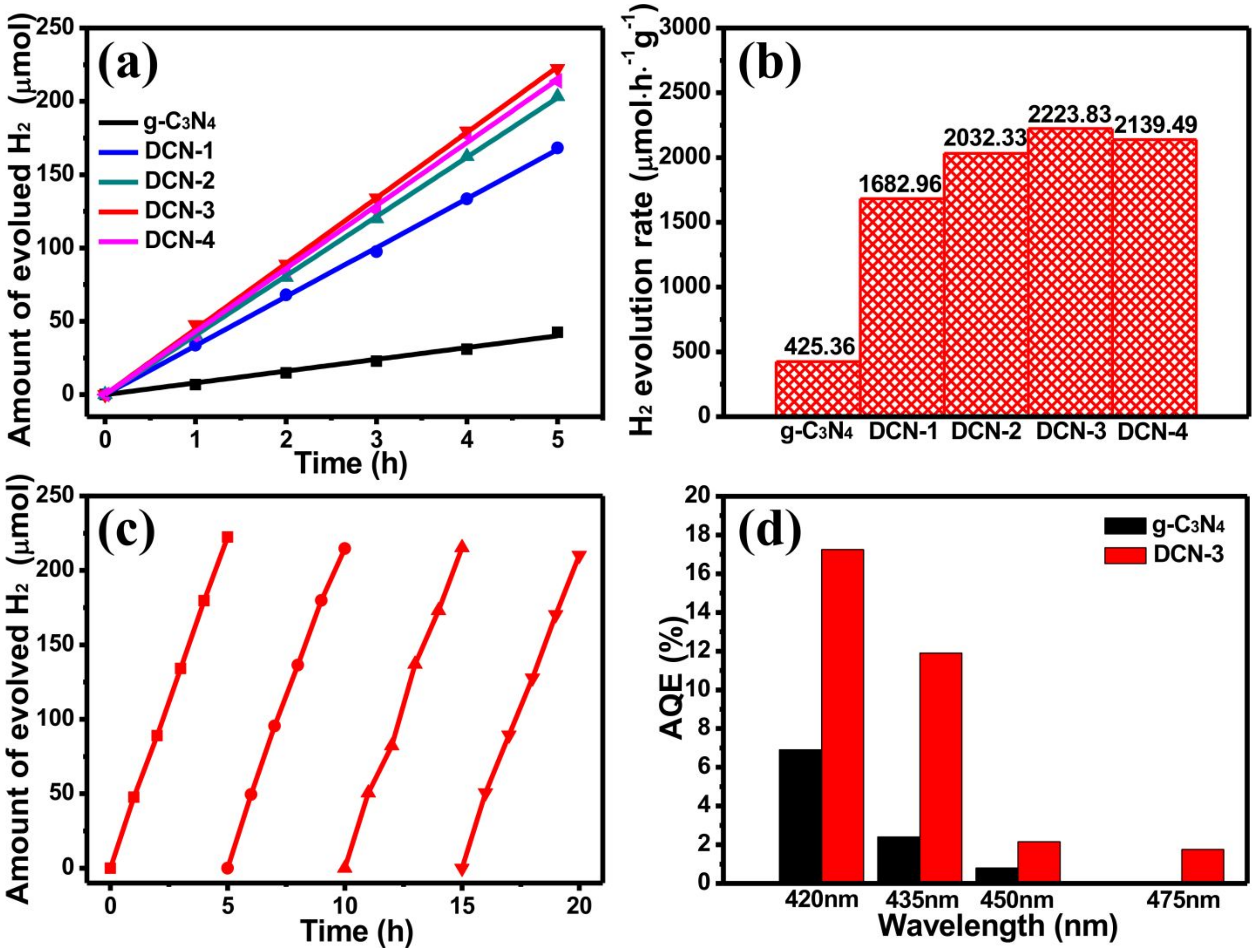

Figure 3. Photocatalytic performance of g-C_3N_4 and defective g-C_3N_4. Amount of evolution H_2 (a) and H_2 production rate (b) of g-C_3N_4, DCN-1, DCN-2, DCN-3 and DCN-4. Cycling test of photocatalytic H_2 evolution for DCN-3. (d) The apparent quantum efficiency (AQE) dependence of g-C_3N_4 and DCN-3 on the wavelength.

In addition, the surface area is one parameter for the photocatalytic activity. Brunauer–Emmett–Teller (BET) surface are measured by N_2 sorption isotherm (as shown in Figure S7). The BET surface areas are 71.4, 53.7 and 20.6 m^2/g for pure g-C_3N_4, DCN-3 and DCN-4, respectively. The decrease of surface area may be the results of the forming aggregates during the treatment and sample wash. Photocatalytic H_2 production activities of the samples were measured under visible light (λ>420 nm) irradiation by using TEOA as the sacrificial agent. The dependence of the H_2 production amount with the time was summarized in Figure 3A. g-C_3N_4 exhibits ~ 42.5 μmol over 5 hours for 20 mg sample. The amount of H_2 production dramatically increases to 168.3 μmol for DCN-1, which is ~ 4 times better than that of pristine g-C_3N_4. The amount of H_2 production can be further increased to 222.4 μmol in the case of DCN-3, which is 5 folders higher than that of pure g-C_3N_4. And then it dropped to ~214.0 μmol. Those indicate that the photocatalytic activity can be enhanced by $NaBH_4$ treatment, which incorporates some defects like -C≡N, B doping into the g-C_3N_4. The normalized H_2 production rates are listed in Figure 3b. The optimal photocatalytic activity can be reached to 2223.8 $\mu molh^{-1}g^{-1}$. To confirm the improved photocatalytic activity comes from $NaBH_4$ treatment, the g-C_3N_4 was treated at 300 ^oC for 30min. the photocatalytic H_2 production amount is obviously lower than pure g-C_3N_4. That further confirms that $NaBH_4$ treatment The DCN also exhibit good photostability for H_2 production. In Figure 3c, the H_2 production amount remains consistent at a prolonged time for 20 hours. No notable decreases of H_2 evolution were detected during the four continuous cycles. It demonstrates DCN can be a stable photocatalyst for H_2 evolution.

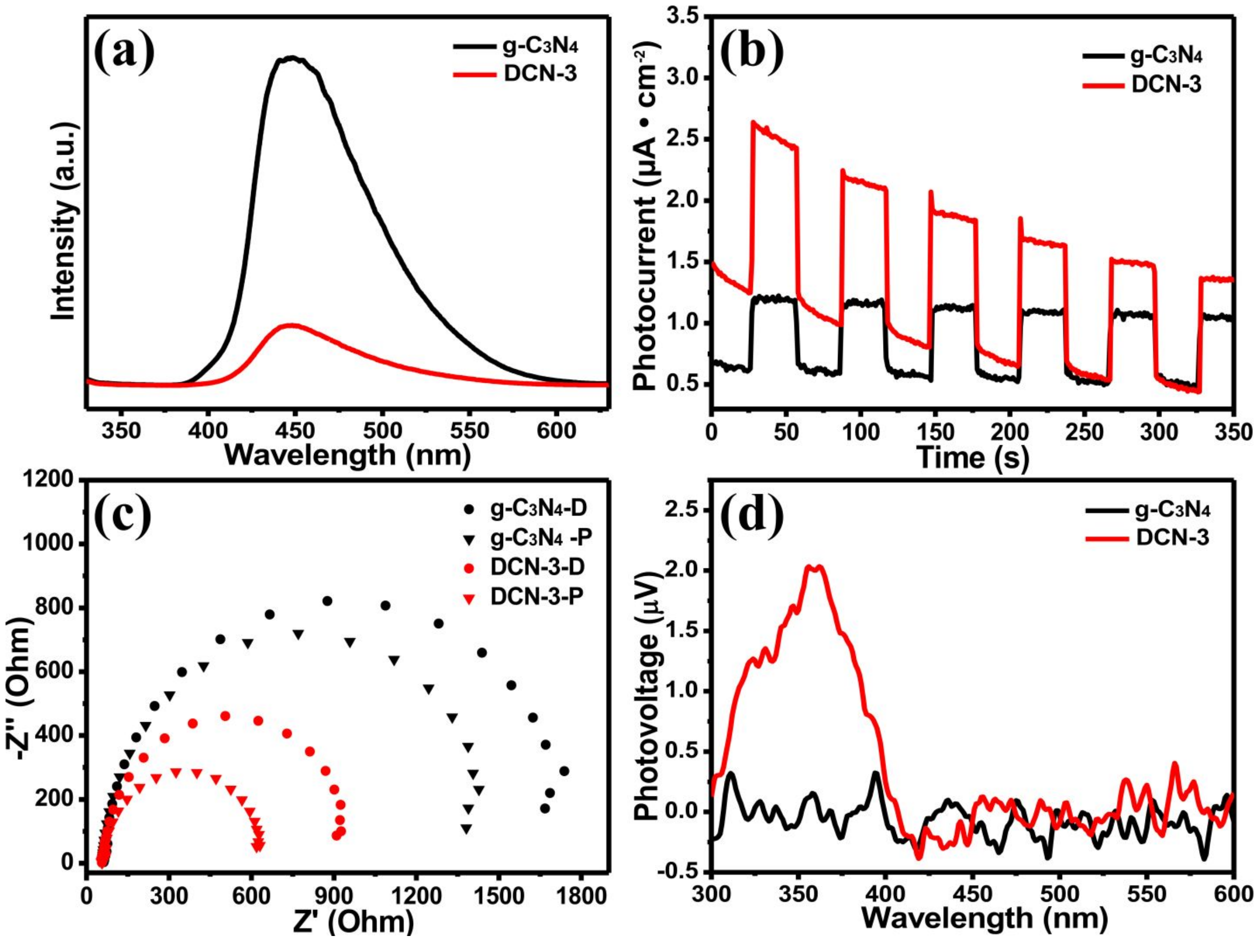

Figure 4. (a) photoluminescence (PL) spectra of g-C_3N_4 and DCN-3. (b) Transient photocurrent density-time plots of g-C_3N_4 and DCN-3. (c) Electrochemical impedance spectroscopy (EIS) of pristine g-C_3N_4 and DCN-3 with/without light irradiation. (d) The surface photovoltage (SPV) response of g-C_3N_4 and DCN-3.

To understand the photocatalytic activity enhancement mechanism, the apparent quantum efficiency (AQE) with different wavelength are investigated. Figure 3d shows the AQE of pristine g-C_3N_4 and DCN-3. The AQE of pure g-C_3N_4 is about 6.90% at 420 ±10 nm, the corresponding the AQE of DCN can reach 17.25%, which is about 2 times higher than that of pure g-C_3N_4. The AQE of g-C_3N_4 dramatically drops to 0.80% at 450 ±10 nm, the DCN remains its AQE of 2.15%

at 450±10 nm. At 475 ±10 nm, no H_2 was collected for g-C_3N_4. The AQE of DCN is still 1.75%. That indicates the g-C_3N_4 has no photocatalytic activity at 475 nm, but DCN keeps its photocatalytic activity. Based on these results, the DCN shows broader visible light response than pure g-C_3N_4. Besides that, the DCN also shows much higher photocatalytic activity at each wavelength than pure g-C_3N_4. That implies the DCN may have better charge separation that pure g-C_3N_4.

To confirm that DCN has higher charge separation efficiency, photoluminescence (PL) spectrum, transient photocurrent, electrochemical impedance, and surface photovoltage were characterized. Figure 4a shows the PL spectrum of g-C_3N_4 and DCN-3. A strong PL emission peak at 450 nm appears in the g-C_3N_4, this peak obviously decreases in the DCN-3. That indicates that radiation recombination is greatly depressed in the DCN-3. When the DCN was excited by the light, the portion of excited electron recombine with the hole at ground state decreases. Thus, the possibility of charge separation increases, that will promote the catalytic reaction happen. The transient photocurrent density indicates the extraction of photogenerated charge from the semiconductor under light irradiation. The transient photocurrent density of g-C_3N_4 and DCN are shown in Figure 4b. The photocurrent density of DCN is about 2 times higher than that of pure g-C_3N_4. It implies that the DCN has higher charge extraction capability than pure g-C_3N_4 and improved charge separation efficiency. That well agrees with the PL results. Furthermore, electrochemical impedance spectroscopy (EIS) of g-C_3N_4 and DCN-3 are displayed in Figure 4c. All of them exhibit semicircles in the middle-frequency region in the case of both dark and light irradiation conditions. The arc radius on the Nyquist plot of DCN is much smaller than that of pure g-C_3N_4, indicating that a smaller charge transfer resistance and a more efficient charge separation process for the DCN-3.[46] The corresponding equivalent circuits are listed in the Figure S4. R_p is the charge transfer resistance across the electrode/electrolyte interface, CPE is the constant phase element for the electrolyte/electrode interface, and R_s is the series resistance. The R_p dramatically decreases from 1710 Ω for g-C_3N_4 to 926 Ω for DCN-3, respectively. This confirms that the DCN-3 has better charge transfer efficiency than g-C_3N_4. Comparing with dark cases, the radius of the semicircle turns smaller with light irradiation. That indicates the light irradiation promotes the charge separation and transfer. Surface photovoltage spectroscopy (SPV) is a powerful tool to characterize charge separation for the photocatalytic system.[47] The powder samples are sandwiched between an ITO electrode and the Cu substrate. Figure 4d shows the SPV response vs. the irradiation wavelength of g-C_3N_4 and DCN-3. It is hard to get the SPV response for the g-C_3N_4 due to the fast recombination of photoexcited charges. An obvious SPV response band in the 300-425 nm is observed, which is related to the electron transition from the VB to the CB of DCN-3. It implies that DCN-3 has high charge separation efficiency than g-C_3N_4. According to the principle of SPS, a positive response is related to the n-type semiconductors due to energy band bending up at the interface between the metal electrode and the semiconductor. And photogenerated electrons tend to flow toward the inside of the semiconductor. SPS results indicate that DCN-3 has better charge separation efficiency. Upon the above results, defects result in broader light adsorption and higher charge separation efficiency in the DCN sample. The defects take the dominant place in the enhancement of photocatalytic H_2 production.

Based on all the results, defects like -C≡N and B dopant was introduced into the g-C_3N_4 during the $NaBH_4$ treatment. The defect changes the valance and conduction band of g-C_3N_4. The valence band XPS as shown in Figure S5, changes from 2.05 eV to 1.92 eV. The band gap also reduces from 2.94 eV to 2.71 eV. Thus, the energy level diagram of g-C_3N_4 and DCN are shown in Figure S5. DCN can adsorb more visible light due to the narrow band gap, DCN has better charge separation efficiency owing to the defect can fast trap the photogenerated electron. That's is why the DCN has high photocatalytic performance.

Conclusions

In summary, DCN samples were prepared by simple $NaBH_4$ treatment at 300 ^oC for 0-60 minutes. The treated samples have a narrow band gap and large visible light response range. XPS results disclose that the B doping happens in the treatment. Photocatalytic H_2 production indicates that DCN sample exhibit higher photocatalytic activity than pristine g-C_3N_4. The AQE illustrate DCN has wide light response range and high photocatalytic activity at a single wavelength. The PL, transient photocurrent, EIS and SPS illustrate DCN sample has better charge separation efficiency than g-C_3N_4. Those are the reasons why DCN samples have high photocatalytic activity.

ASSOCIATED CONTENT

Supporting Information.

The following files are available free of charge.

More the schematic synthesis route, SEM images, equivalent circuit of EIS spectra, VB XPS and band alignment of g-C_3N_4 and DCN-3 (PDF file)

ACKNOWLEDGMENT

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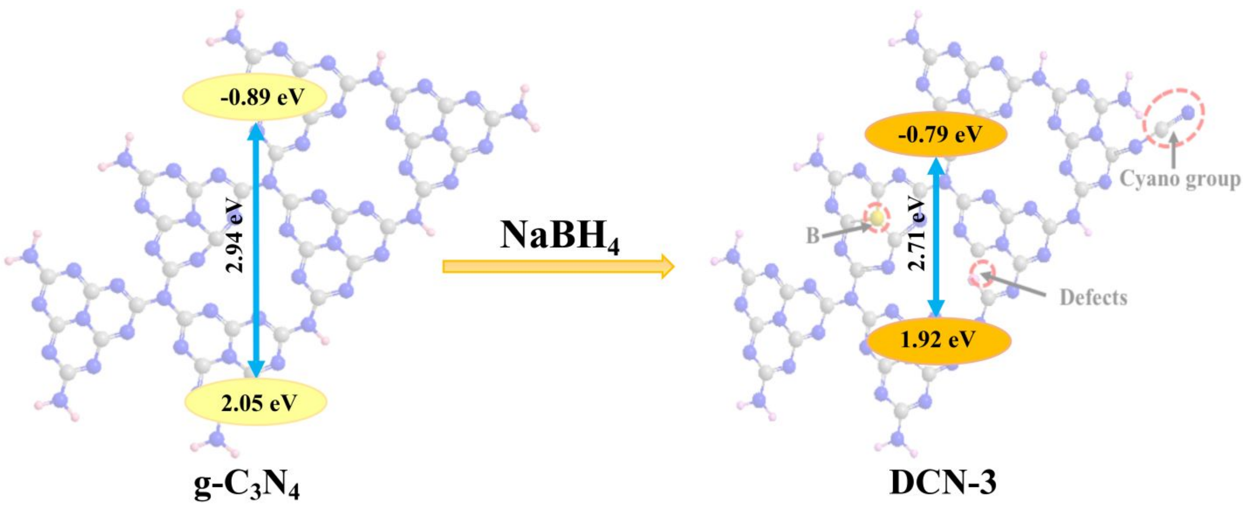
Graphtic abstract



The defects (B, -C≡N and dangling bond) are introduced into g-C_3N_4 through the treatment of $NaBH_4$ resulting in a narrow bandgap and efficient charge separation efficiency. As a consequence, the photocatalytic performance is greatly promoted.